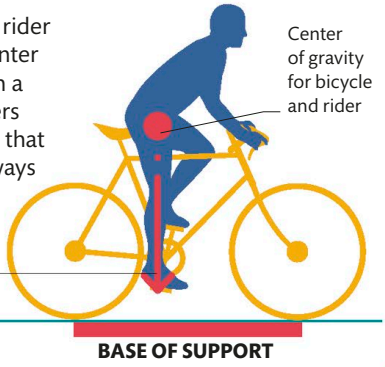
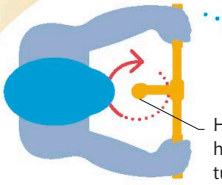
MAINTAINING BALANCE

To balance on a bicycle, a rider must control his or her center of gravity. When cycling in a straight line, the rider steers toward a fall to make sure that the center of gravity is always over the wheels, forming the base of support.



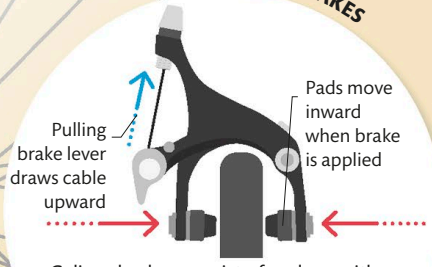
HEADSET

HANDLEBARS



Handlebars are levers that magnify the input force to turn the front wheel. Some bicycles have drop handlebars. These make the rider bend lower, into a more aerodynamic position.

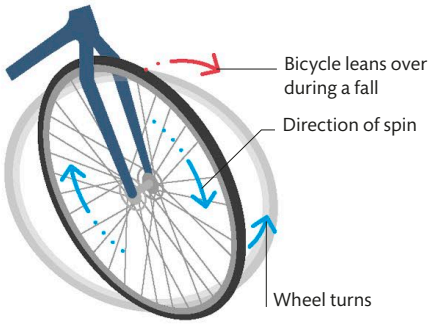
BRAKES



Caliper brakes consist of pads on either side of each wheel. Squeezing the brake lever pulls a cable that makes the pads grip the wheel, increasing friction and slowing the wheel.

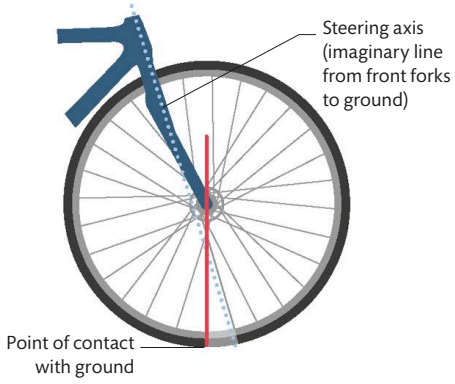
Freewheeling

Two mechanical principles usually help to explain why a bicycle can stay upright: the gyroscopic effect and the caster effect. Recent research suggests another important influence is the fact that the front of the bicycle has a center of gravity that is lower than the rear and forward of the steering axis. During a fall, the front of the bicycle falls faster than the rear, turning the front wheel toward the fall and keeping the bicycle upright.



THE GYROSCOPIC EFFECT

The front wheel acts like a gyroscope. If the bicycle falls to one side, the gyroscopic effect turns the wheel to the same side, keeping the bicycle upright.



THE CASTER EFFECT

The point at which the front wheel meets the ground trails behind the steering axis, like a caster on a trolley. This means that the wheel always turns in the same direction as the bicycle is traveling.